UNIT – 3

ASSESSMENT – 1

PROBLEMS

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1.Smart Medical Diagnosis Using Forward and Backward Chaining

Problem Statement

A hospital uses a rule-based logical agent to perform preliminary diagnosis. The system contains rules relating symptoms such as fever, cough, rash and breathlessness to possible diseases. Patient A has fever, cough and breathlessness. The objective is to use **Forward Chaining** to derive all possible diagnoses and **Backward Chaining** to verify whether Patient A has pneumonia.

Knowledge Base

- Fever AND Cough → Possible Flu
- Fever AND Rash → Possible Measles
- Cough AND Breathlessness → Possible Pneumonia
- Patient A has Fever
- Patient A has Cough
- Patient A has Breathlessness

2. **Autonomous Traffic Management Using First-Order Logic**

Problem Statement

A smart-city traffic controller uses a First-Order Logic knowledge base to manage emergency vehicles. Emergency vehicles are given a clear path and green signal. Vehicles behind an emergency vehicle can proceed when that vehicle receives a green signal. The objective is to use unification and resolution to prove that CarA can proceed.

3. **Agricultural Expert Reasoning Using Resolution**

Problem Statement

An agricultural advisory system determines whether drip irrigation should be applied based on soil and crop conditions. If the soil is dry, irrigation is needed. If irrigation is needed and the crop is wheat, the drip method is applied. The objective is to convert the knowledge base into CNF and use resolution refutation to prove that the drip method should be applied.